Exp - 8

N-MAP to Discover Live Hosts Using N-Map scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform

Aim :-

To discover live hosts on the network using Nmap host discovery scans - ARP, ICMP, TCP/UDP Ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies

Introduction

*Nmap ("Network Mapper") is a free, open-source network scanning tool created by Gordon Lyon

*) It is the industry standard tool for Mapping networks, Discovering live host

*) This lab focuses on first of two questions asked during reconnaissance :

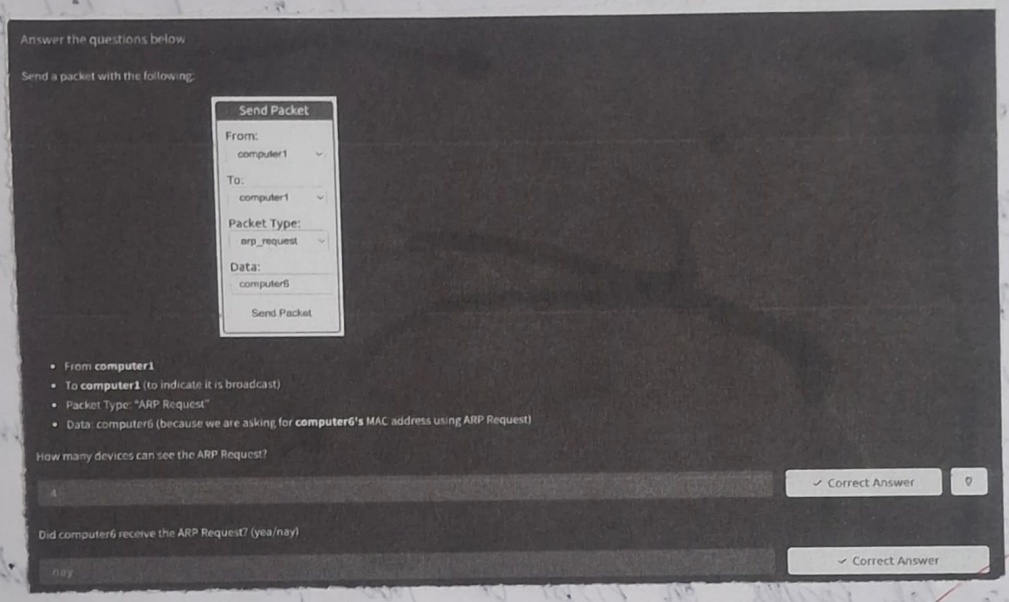
1. Which System are up? answered in this room
2. What services are running on those systems? - answered in the follow-on Nmap port Scanning rooms

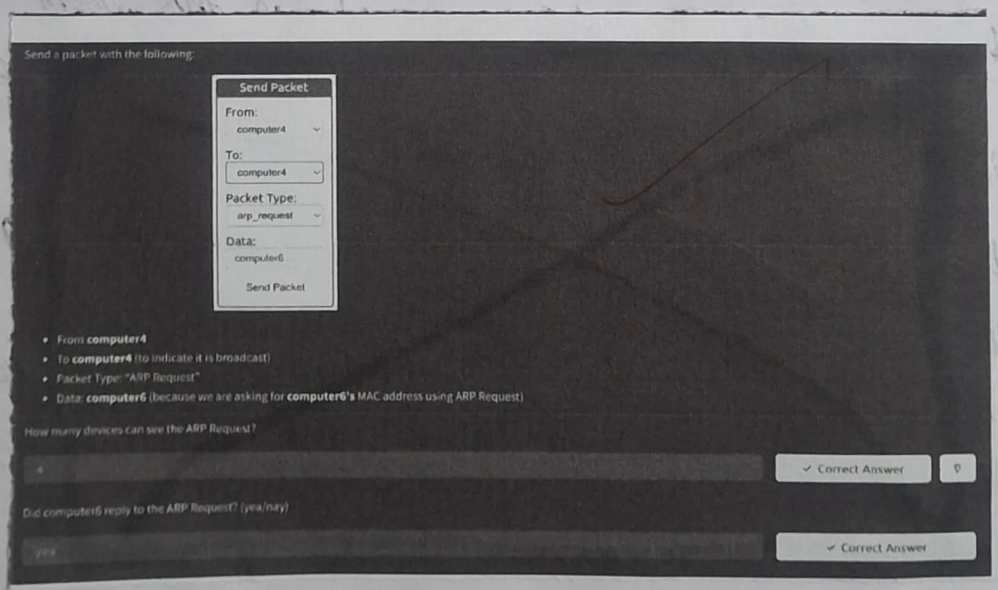
Nmap can determine whether a host is alive using four different protocols,

* ARP (Link Layer)
* ICMP (Network Layer)
* TCPC (Transport Layer)
* UDP (Transport Layer)

ARP Host Discovery

* On a local Ethernet, communication requires the destination MAC address
* To learn it, a host broadcasts an ARP any host holding that IP replies with MAC address
* A reply proves the host is up.
* ARP only works within your own screenshot.





ICMP Host Discovery

* Request Several request/Replay pairs that can reveal a line host
*) Windows System & many Firewalls blocks ICMP

7.1 ICMP Echo (-PE)

$ Sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.0.1
Host is up (0.069s latency)
Nmap Scan report for 10.200.6.50
Host is up
Nmap done: 256 IP address (3 host) scanned in 3.38 seconds.

ICMP Timestamp (-PP)

$nmap -PP -sn 10.200.6.01.24
Nmap Scan report for 10.200.6.50
Host is VP
Nmap done: 256 IP address (2 host up) scanned in 8.42 seconds

ICMP address mask (-PM)

```
$nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP address scanned in 52.14
          second
```

8.1 TCP SYN PIN (-PS)

Send a TCP Packet with the SYN flag to target port. An oper port answers SYN/ACK; a closed port answers RST - either reply confirms the host is up

```
$ nmap -PS -sn 10.200.6.0/24
Nmap Scan report For 10.200.6.50
Host is up
Nmap Scan report For 10.200.6.250
Host is up
Nmap done: 256 IP address
```

TCP ACK pind (-PA)

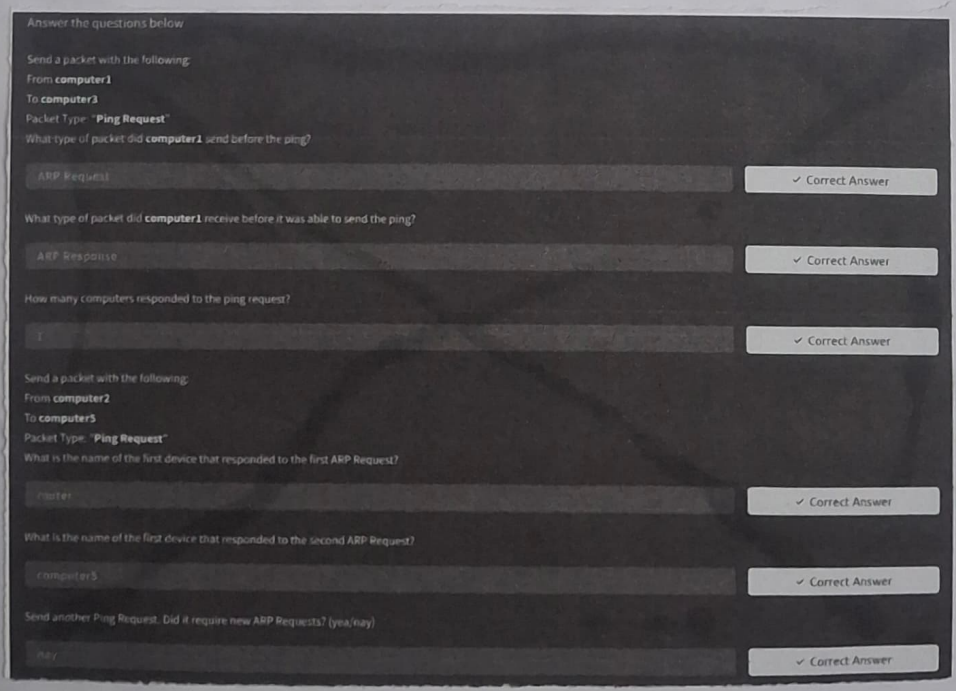
Sends a TCP packet with the ACK flag set. Because the ACK does not belong to any existing connection.

$sudo nmap -PA -sn 10.200.6.0
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.068s latency)
Nmap done: 256 IP addresse (2 host up) scanned in 33.88 seconds

UDP Ping (-PU)

Sends a UDP packet to a UDP port.

$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.064s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 35.97 seconds.



Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP ✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

-PM ✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE ✓ Correct Answer

Answer the questions below

Which TCP ping scan does not require a privileged account?

TCP SYN Ping ✓ Correct Answer

Which TCP ping scan requires a privileged account?

TCP ACK Ping ✓ Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

-PS23 ✓ Correct Answer

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

10.10.12.8 ✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

6400 ✓ Correct Answer

Answer the questions below

We want Nmap to issue a reverse DNS lookup for all the possible hosts on a subnet, hoping to get some insights from the names. What option should we add?

-R ✓ Correct Answer

Answer the questions below

How many hosts are found alive after scanning the CONNECTION_IP/24?

1 ✓ Correct Answer

Result :-

* Nmap Discover live host using nmap has been done successfully